

Bachelor AI - MLOps

21 FREE EXERCISES

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Suggestions for Free Exercises

Below you will find a list of possible exercises related to what you have learned. Choose those that interest you the most.

Extend the Mushroom Example

We have not yet implemented some interesting aspects of the Mushroom model.

- Add a Real-time (RT) feature. To do this, add a simple additional feature in the training code that is calculated from existing features. For example, an average. Also adjust the inference code and data generator accordingly.
- Use another additional feature that was pre-calculated in a batch process. For example, a mushroom identification app could use the current coordinates to look up in a table which mushrooms are typical for the region. In terms of approach, this is not much different from the first suggestion.
- It becomes more difficult if the mushrooms typical for the region change frequently. Perhaps we work in a nuclear-contaminated environment and catalog mushrooms together with many other people, and need the average mushroom size of the last 30 minutes in the region. We would have to calculate this as a Near Real-time (NRT) feature in a streaming process.

We have only operationalized a very simple model in the course. It would be interesting to also operationalize a model that performs feature calculations and has a natural label, thereby allowing more monitoring and retraining. Here are three ideas with data sources:

- Build a prediction for **air quality**
- Predict the **Bitcoin price**
- Make a **weather forecast**

Would you like to learn more about Feature Stores and use a Feature Store? Then Hopsworks is a good resource. The Hopsworks Feature Store can be used well for experiments with a free account. Hopsworks offers a large number of **Tutorials** as Jupyter notebooks. Some suggestions:

- The most important concepts (**Feature Store**, **Feature Groups**, **Feature Views**)
- **Quickstart notebook to get started**
 - **Variant of the Fraud Batch Tutorial with Data Validation with Great Expectations**
- Complete use cases: **Batch AI Systems** and **Real-time AI Systems**
- Advanced: **LLM AI Systems** and **API Examples**

If you are interested in Data Drift, you will find a lot of good material on the Evidently website.

- Tutorials with Notebooks
- Blog Posts, for example
 - How to handle ML model drift in production
 - How to interpret data and prediction drift together?
 - 5 methods to detect drift in ML embeddings

Weights and Biases (short wandb) is a cloud service that offers, among other things, **Experiment Tracking** and a **Model Registry**.

- Work through the **Quickstart** and **Get Started with W&B Models**, browse the **Example Notebooks** and read the **Documentation** to get to know the product
- Replace MLFlow in our infrastructure with wandb
- Rewrite the existing code to use wandb instead of MLFlow

If you want to work with orchestration, Apache Airflow is a tool you can try. You can find a tutorial with code here:

- Why use Apache Airflow (or any orchestrator)

If you want to know more about Great Expectations, you can check out the following suggestions.

- [A short tutorial](#) with code on using Great Expectations in data pipelines
- [A short article](#) on Data Quality Checks

Would you like to get to know Grafana better and expand monitoring? Here are some ideas.

- Check out the [Grafana Quickstart](#)
- Extend the drift detection dashboard with visualizations that
 - show the progression of calculated distances
 - display the state of the dataset well in a single visualization
 - Set a threshold and have Grafana display an alarm as soon as it is exceeded
- Get inspired by the [Visualizations of Evidently](#)
- Build a dashboard to monitor the operation of Statsd, Prometheus and Grafana (but there are many of those that you can simply download)

If you want to learn more about Stream Processing, check out the short tutorial below.

- [Data Engineering Project: Stream Edition](#)

If you want to learn more about Docker, the following longer tutorial is well suited to acquire all the necessary basics.

- Docker Curriculum
- After that, you can try to containerize the Data Generator

Would you like to know how larger companies implement MLOps?

- Read some **Case Studies**

If you simply want to browse through various articles, this is a good list to start with:

- [MLOps Guide by Chip Huyen](#)